UCONN FALL FEST 2026 · IBM QISKIT × qBRAID CHALLENGE

Instant Insanity on a Quantum Computer

Constrained p = 12 QAOA with W-states and XY mixers, checked against brute force at every step

Team Phase Flip

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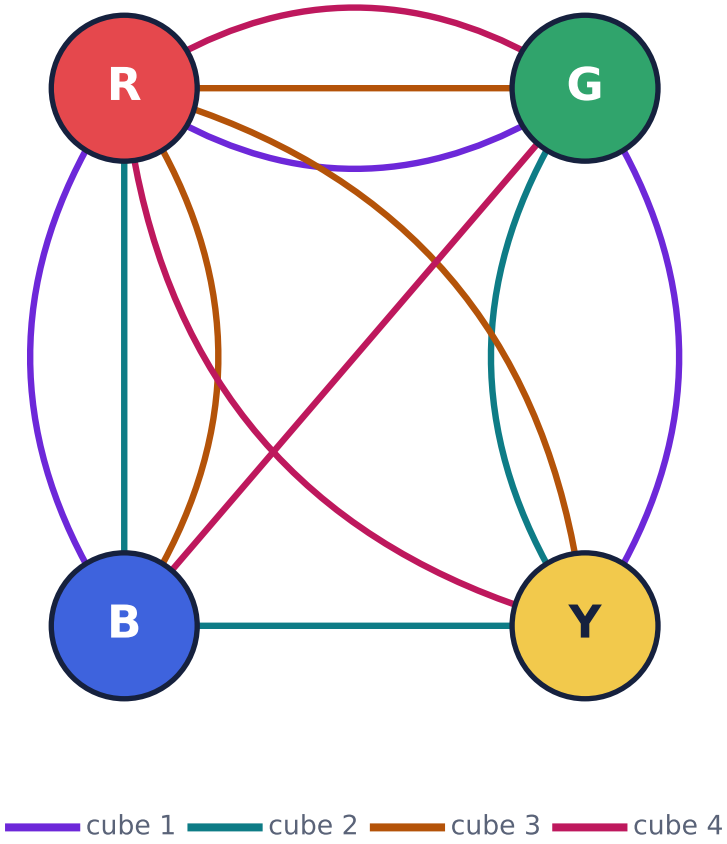
Levels reached: Warm-up ✓ Core (4 cubes) ✓ Bonus (N = 5 to 8 cubes) ✓

github.com/scarthur/IBM-Qiskit-Team-Phase-Flip

The puzzle: no colour may repeat on any side

- Each cube has 3 pairs of opposite faces in 4 colours (R, G, B, Y).
- Stack all 4 cubes: front, back, left and right must each show every colour exactly once.
- Each cube has 24 orientations: $24^4 = 331{,}776$ towers to try, and only 8 of them work (one tower, 8 symmetric views).
- Graph view: one node per colour, one edge per opposite pair.
- One axis (e.g. front/back) = one edge per cube, every colour touched twice: a 2-factor.
- Solution = two 2-factors that never use the same pair of a cube. Of $6^4 = 1{,}296$ graph cases, exactly 2 solve it.

Colour graph of our 4 cubes



Encoding: one-hot picks of opposite-face pairs

- x[c][p] = 1 if cube c puts pair p on front/back; y[c][p] for left/right.
- 6 qubits per cube → 24 qubits for the full tower (qubit 6c+p = x, 6c+3+p = y).
- Warm-up and bonus: front/back only, 3 qubits per cube (3N qubits).
- Cost (QUBO, zero exactly for a solved tower):

$$C = \sum_{\text{axis } k}\sum (d_k - 2)^2 \; + \; \mu \sum_{c,p} x_{cp}\, y_{cp} \; + \; \lambda \sum_{c} \Big[\big(\sum_{p} x_{cp} - 1\big)^2 + \big(\sum_{p} y_{cp} - 1\big)^2 \Big]$$

- (d_k − 2)²: every colour k must appear exactly twice on each axis (the 2-factor rule).
- μ·x·y: a pair can't serve both axes. λ: one pair per axis. We use λ = μ = 2.
- Mapped to Pauli Z / ZZ terms with x = (1 − Z)/2. In the circuit the one-hot λ term is dropped:

  the W-state + XY mixer never leaves the one-hot subspace.

OUR QUANTUM APPROACH

Algorithm: constrained QAOA with p = 12 layers

Each cube starts in a W-state (|100⟩+|010⟩+|001⟩)/√3 and is mixed with XX+YY gates, so every shot is a valid one-hot pick.

p = 12 (cost, mixer) layers → 24 angles per circuit, optimised separately for every circuit and puzzle (SEED = 7).

Training the angles

Exact simulator on the one-hot states only: 81 at 12 qubits, 6,561 at 24 (not 16.8 M).

Adjoint gradient: all 24 derivatives in one pass.

L-BFGS-B maximises P(lowest cost) directly, from 6 annealing-like ramps.

Trimming the circuit

One-hot reduction: inside a cube's group ZZ = Z + Z − 1, saving 2 CX per term.
12q: 1,640 → 1,424 CX
24q: 3,568 → 3,136 CX

rzz gates packed into rounds on disjoint qubits: same unitary, lower depth.

Three ways to use it

A · 24 qubits: the whole tower in one circuit.

B · 12q + 4q: front/back QAOA, then a 4-qubit QAOA picks left/right (quantum both stages).

Bonus · 3N qubits: front/back QAOA, then classical pairing of sampled 2-factors.

Classical checks before claims

✓ **Hamiltonian = classical cost on all 1,296 choices**

max |H − cost| = 0; exactly 2 zero-energy states, both pass solves()

✓ **Our fast simulator = Qiskit Statevector**

real 12-qubit p = 12 circuit: max |ΔP| = 4.9e-16, leakage 7.5e-15

✓ **Adjoint gradient = finite differences**

max error 2.3e-09; one-hot reduction leaves every energy unchanged

✓ **Shots match the exact prediction**

warm-up 89.9% vs 91.1%; 24-qubit tower 19.55% vs 18.45%

✓ **Every reported tower re-checked**

independent solves() on every shot we count; decoder uses Qiskit order

✓ **Bonus puzzles have exactly one tower**

seeds {5:16, 6:86, 7:68, 8:4}, confirmed by brute force

Verified tower (both methods)

	Front	Back	Left	Right
cube 1	G	Y	B	R
cube 2	Y	B	Y	G
cube 3	B	R	R	Y
cube 4	R	G	G	B

Each column shows R, G, B, Y once

Honest note: brute force solves the 4-cube puzzle in under a second.

Results: 4-cube puzzle, p = 12 (local AerSimulator, SEED = 7)

Experiment	Qubits	CX	Depth	Shots	Measured	Random guess	Gain
Warm-up: any front/back 2-factor	12	1,424	781	1000	**89.9%**	4.94% (4/81)	18×
Method A: full solved tower	24	3,136	820	2000	**19.55%**	0.0305% (2/6561)	641×
Method B stage 1: 2-factor	12	1,424	781	1000	**89.9%**	4.94% (4/81)	18×
Method B stage 2: tower | FB	4	72	136	1000	**92.1%**	6.25% (1/16)	15×

p = 2 → p = 12

Circuit	Success p = 2	Success p = 12	CX p = 2 → 12
Warm-up 2-factor (12q)	28.2%	**89.9%**	300 → 1,424
Method A tower (24q)	1.60%	**19.55%**	648 → 3,136
Method B stage 2 (4q)	26.2%	**92.1%**	12 → 72

What it tells us

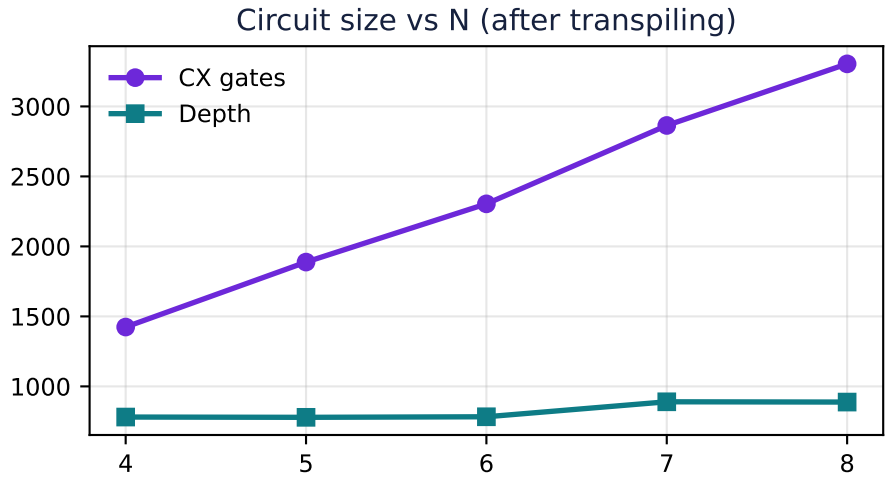
More layers + angles trained per circuit for the success rate itself lift every result 3 to 12×, at about 5× the gate count.

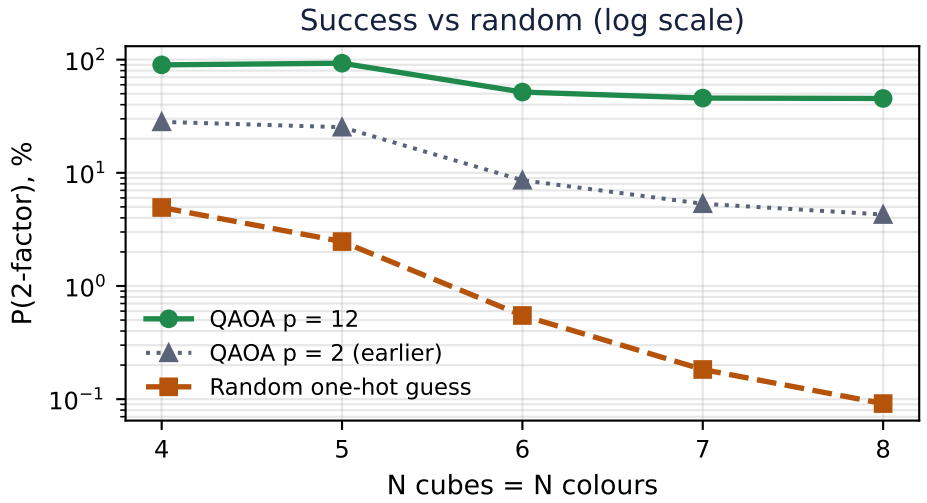
Splitting the tower (Method B) still beats one 24-qubit circuit.

SCALING

Bonus: the same method scales to N = 8 (p = 12)

N (seed)	Qubits	CX	Depth	P(2-fac.)	Random	Gain	Tower
4 (CUBES)	12	1,424	781	**89.9%**	4.94%	18×	✓
5 (16)	15	1,888	779	**93.0%**	2.47%	38×	✓
6 (86)	18	2,304	783	**51.7%**	0.55%	94×	✓
7 (68)	21	2,864	890	**45.8%**	0.18%	250×	✓
8 (4)	24	3,304	888	**45.4%**	0.091%	496×	✓





- Random = number of 2-factors / 3^N (brute-force count).
- CX grows about linearly (≈ 470 per cube); depth stays near 800 because rzz gates run in parallel rounds.
- Success ≥ 45% up to N = 8; lead over random grows from 18× to 496×. A solves()-checked tower was found for every N by pairing sampled 2-factors.

TAKEAWAYS

What we learned, and what's next

What we can and can't claim

- ✓ The model is exact: H matches brute force on all 1,296 cases.
- ✓ p = 12 lifts the 24-qubit tower 1.6% → 19.6% and the 2-factor rate 28% → 90%.
- ✓ 18× to 641× above random; every tower passes solves().

- ✗ No speedup: angle training enumerates every one-hot candidate classically, so it already knows the answer. Brute force is instant.
- ✗ Local AerSimulator only (seed 7); SV1 path is wired in (USE_SV1 = True), no job IDs yet.
- ✗ 1,400–3,300 CX at depth ~800: too deep for today's noisy hardware.

One thing we'd try next

Train the angles without knowing the answer: optimise ⟨cost⟩ or CVaR (the best shots only) on small puzzles, then transfer the same 24 angles to larger N, where we can't enumerate.

Then cut depth for hardware: fewer layers, a mixer over each cube's 6 valid choices, or 3 qubits per cube (compact encoding).

Run the final circuits on SV1, then compare with a QPU using earned credits.